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MATHEMATICS
Paper 1
Jul/Aug 2026
2¼ Hours

JINJA JOINT EXAMINATIONS BOARD

Uganda Certificate of Lower Secondary Education

MOCK EXAMINATIONS - JULY / AUGUST 2026

MATHEMATICS

2 hours 15 minutes

INSTRUCTIONS TO CANDIDATES;

This paper consists of two sections; **A** *and* **B**. *It has* **six** *examination items.*
Section **A** *has* **two** *compulsory items.*
Section **B** *has* **two** *parts;* **I** *and* **II**. *Answer* **one** *item from each part.*
Answer **four** *(4) examination items in all*
Any additional item(s) answered will **not** *be scored.*
All *answers* **must** *be written in the answer booklet(s) provided.*
Graph paper will be provided
Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

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SECTION A

Answer **all** *items in this section*

ITEM 1

At birthday party of your uncle James on Monday 10th may 2025, Him and his two sons agreed to startup a hardware business which required eight million, fifty thousand shillings. They planned to contribute for this cause as follows;

Uncle James: 60% of the total money required, 0.25 of the remaining amount to be contributed by the young son while the $\frac{2}{3}$ of the balance for the old son. The remaining

balance was to be borrowed from Uncle James' friend who accepted to send the young son to withdraw the money from his account using the ATM card. For security reasons, he wrote his pin as 21 20 21 13 all in a number system less than five which has to be converted to a decimal base to get the actual pin.

They agreed to bring money and startup their business when the two sons meet home with their father. The older son comes home after every 16 days and the young son comes after every 12 days. Uncle James wanted to know when his sons are meeting home again for proper preparations for the business startup.

Task;

a) Determine how much each contributed towards the start of the business.

b) Determine the actual pin used to withdraw money from the friend's account.

c) Determine when the business started.

ITEM 2.

A transport company is preparing to transport 700 passengers using 10 drivers and its fleet of buses and coasters. Each bus can carry 70 people at a time while each coaster carries 35 people at a time. The manager wants to take the required vehicles for service and fueling to avoid any inconveniences that may arise.

This company also owns a truck which the contractor is planning to use to deliver construction materials to the site. This truck can carry 3000kg of cargo or more. The contractor needs more bags of cement than bags of lime.

A bag of cement weighs 50kg and costs Sh. 25000 while a bag of lime weighs 80kg and costs Sh. 20,000. A contractor has a budget of 1.5 million for buying materials.

Task;

a) How many of each vehicle type should be worked on?

b) Determine the maximum number of bags of cement gravel bags he should buy in order to meet his budget.

SECTION B

This section has two parts; **I** *and* **II**

Part I

Answer **one** *item from this part*

ITEM 3.

A survey was carried out in preparation for s.4 leavers' party about which drinks they preferred. Out of **60** learners surveyed, **34** preferred juice and **28** preferred yoghurt and they are some other learners who preferred sodas though **no** learner preferred soda only. The number of learners who preferred soda and juice only are **equal to** those who didn't prefer any of the 3 drinks, and those who preferred soda and yoghurt only are **four times** those who preferred yoghurt only. **9** learners preferred juice and yoghurt only while **4** preferred all the three drinks.

In order to make the party colorful and enjoyable, every learner took what s/he preferred. These drinks are sold in cartons of 6 bottles each and each carton of soda, juice and yoghurt costs Sh. 11,000, Sh. 17,500 and Sh. 19,000 respectively.

Task;

a) Determine the probability that a learner preferred soda.

b) Determine the minimum number of cartons of each type of drinks that will be purchased such that learners enjoy soft drink.

c) Using the knowledge of matrices, determine the minimal amount of money required for drinks.

ITEM 4

A mini bus travelled from town A to town B which is 360km apart. It begun the journey at 7: 00 am moving at a steady speed of $50kmhr^{-1}$ for 2 hours, after which it got a mechanical problem. It took them 60 minutes to fix it after which it then travelled the remaining journey to town B in 3hours.

A motorist started following a minibus at 8:30am for a passenger in the minibus forgot her bag behind. He travelled at a steady speed of $20kmhr^{-1}$ more that of the minibus in order to catch up with the passenger.

Meanwhile, a taxi left town B at 8: 00am heading towards town A, driving nonstop at a steady speed and completed the journey in 4 ½ hours.

Task;

a) Determine the difference in times of arrival at these vehicles' destination.

b) Determine when and where the two vehicles met.

c) With a reason, determine whether the motorist was able to catch up with the minibus for the passenger's bag.

d) Find the average speed for a minibus for the whole journey.

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Part II

Answer **one** *item from this part.*

ITEM 5

A tourist from United States converted his \$12,000 to Ugandan shillings at rate of \$1 = UGX 3560. He rented in a hotel that charges UGX 20,000 daily a month. He did this because he wanted to startup a business in Uganda. At this business, he plans to employ 3 people of which he will be paying \$150 each including UGX.280,000 nontaxable allowance per month. He will pay them according to the tax structure below;

Taxable income (UGX)	Tax Rate (%)
0 – 100,000	Free
100,001 – 200,000	5
200,001 – 350,000	10
350,001 – 500,000	18
Above 500,000	25

From the hotel at the border, the tourist plans to begin his business 150km away in town P on a bearing of $S30^0E$ and he will be supplying some of his products in a town Q which is 210km away from town P but due East of the hotel at the border. To save time and fuel, he wants to use a direct route from town Q to the hotel but is not sure of how long it is. **Given that a month has 30 days.**

Task;

a) Determine the amount of money the tourist is planning to start the business with in Ugandan shillings after paying the monthly rent.

b) How much do these employees receive altogether monthly?

c) Determine the actual distance between the hotel and town Q.

ITEM 6

Mr. Kato a householder is facing a problem of water shortages. To solve this problem, he plans to harvest rain water and he has contacted the tank making company for a tank with a rectangular cross-section of height 3m and the base 2m by 1.5m.

The upper part of the tank is to have a small circular inlet of radius 7.5cm where the pipe from the roof will be joined to the tank while the lower end is to have a small circular hole of diameter 4cm where the tap will be placed for easy fetching of water from the tank. The company manager realized that the material available to make a tank is only $30.51m^2$ but is not sure whether it is enough to meet Mr. Kato's order.

While standing at the top of the tank, Mr. Kato with a height of 1.5m can see his flower garden at an angle of depression of 58^0.

Task;

a) Determine whether this company is able to make this tank using the available material or not.

b) Find how far is the flower garden from the bottom of the tank.

c) If your neighbor needs a small water tank whose material is of area $20m^2$ similar to that of Mr. Kato, determine the actual dimension of the base of this smaller water tank.

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